$$\alpha_1 + \alpha_2 + \beta_5 \leqslant 1 \tag{17.76}$$

$$\alpha_1,\ \alpha_2,\ \alpha_3 \quad \text{free} \tag{17.77}$$

$$\beta_1,\ \beta_2,\ \beta_3,\ \beta_4,\ \beta_5 \leqslant 0 \tag{17.78}$$

求解上述模型，与前 2 次的无界不同的是，这次我们终于得到了一个最优解，该最优解对应的极点坐标（也就是最优解的取值）为

$$\text{extreme point} = \left[1,\ 1,\ 1,\ 0,\ 0,\ 0,\ -1,\ -1\right]^{\mathrm{T}}$$

$$\text{obj} = q^*(\bar{y}) = 8$$

并且注意到，当前子问题 (SP($\bar{y}$ fixed)-Dual iter 3) 的目标函数 obj $= q^*(\bar{y}) = 8$，与前一次迭代中主问题 (Master Problem Iter 3) 得到的 $\bar{q} = 0$ 并不相等，因此我们没有得到最优解，继续迭代算法。我们根据极点的坐标，构造出一个 Benders Optimality Cut，其数学表达式如下：

$$\left(\boldsymbol{\alpha}_p^j\right)^{\mathrm{T}}(\boldsymbol{b} - \boldsymbol{B}\boldsymbol{y}) \leqslant q$$

将 $\boldsymbol{\alpha}_p^j$ 和 $(\boldsymbol{b} - \boldsymbol{B}\boldsymbol{y})^{\mathrm{T}}$ 都代入，得

$$(\boldsymbol{b} - \boldsymbol{B}\boldsymbol{y})^{\mathrm{T}}\boldsymbol{\alpha}_p^j = \left[8,\ 3,\ 5,\ 8y_1,\ 3y_2,\ 5y_3,\ 5y_4,\ 3y_5\right] \cdot \begin{bmatrix} 1 \\ 1 \\ 1 \\ 0 \\ 0 \\ 0 \\ -1 \\ -1 \end{bmatrix} \leqslant q$$

整理得

$$16 - 5y_4 - 3y_5 \leqslant q$$

17.2.5 第 4 次迭代

再次更新松弛主问题。在 (Master Problem Iter 3) 中添加上一次迭代产生的 Benders Optimality Cut，更新后的主问题变为

$$\begin{aligned} \min\ & 7y_1 + 7y_2 + 7y_3 + 7y_4 + 7y_5 + q && \text{(Master Problem Iter 4)} \\ \text{s.t.}\ & 16y_1 + 5y_4 + 3y_5 \geqslant 8 && (17.79) \\ & 5y_3 + 5y_4 \geqslant 5 && (17.80) \end{aligned}$$

$$5y_4+3y_5 \geqslant 16-q \tag{17.81}$$

$$q \geqslant 0 \tag{17.82}$$

$$y_1,\quad y_2,\quad y_3,\quad y_4,\quad y_5 \quad \in \{0,1\} \tag{17.83}$$

上述模型 (Master Problem Iter 4) 可以得到一个最优解：

$$\bar{y} = \begin{bmatrix} 0, & 0, & 0, & 1, & 1 \end{bmatrix}^{\mathrm{T}}$$

$$\text{obj} = q^*(\bar{y}) = 8$$

注意到该解 $\bar{y}$ 与上一次迭代之后的值相同，且 $\text{obj} = q^*(\bar{y}) = 8 = \bar{q}$。因此，算法停止，得到了最优解（注意，算法终止的条件是 $q^*(\bar{y}) = \bar{q}$）。

17.3 Benders 分解应用案例

17.3.1 固定费用运输问题

本节我们以固定费用运输问题（Fixed Charge Transportation Problem, FCTP）为例来介绍 Benders 分解算法的应用（本节案例来自参考文献 Kalvelagen，2002）。参照第 2 章的内容，我们写出运输问题的数学模型如下：

$$\min_{\boldsymbol{x}} \quad \sum_{(i,j)\in A} c_{ij}x_{ij} \tag{TP}$$

$$\text{s.t.} \quad \sum_j x_{ij} = s_i, \quad \forall i \in S \tag{17.84}$$

$$\sum_i x_{ij} = d_j, \quad \forall j \in C \tag{17.85}$$

$$x_{ij} \geqslant 0, \qquad \forall (i,j) \in A \tag{17.86}$$

在第 2 章我们介绍到，固定费用运输问题是运输问题的一个拓展，在固定费用运输问题中，我们考虑两点之间的设施是有固定费用的，即对于弧段 $(i,j) \in A$，如果在弧段上运输，则会产生一次性的固定成本 f_{ij}。因此我们需要引入决策变量 y_{ij} 标识弧段 $(i,j) \in A$ 是否被启用或者关闭。根据第 2 章的内容，我们给出固定费用运输问题的数学模型如下：

$$\min_{\boldsymbol{x},\boldsymbol{y}} \quad \sum_{(i,j)\in A} (f_{ij}y_{ij} + c_{ij}x_{ij}) \tag{FCTP}$$

$$\text{s.t.} \quad \sum_j x_{ij} = s_i, \qquad \forall i \in S \tag{17.87}$$

$$\sum_i x_{ij} = d_j, \qquad \forall j \in C \tag{17.88}$$

$$x_{ij} \leqslant M_{ij}y_{ij} \tag{17.89}$$

$$x_{ij} \geqslant 0, y_{ij} \in \{0,1\}, \quad \forall (i,j) \in A \tag{17.90}$$

其中，M_{ij} 是足够大的正数，这里取

$$M_{ij} = \min\{s_i, d_j\}$$

上述模型可以重写为

$$\min_{x,y} \quad \sum_{(i,j)\in A} (f_{ij}y_{ij} + c_{ij}x_{ij}) \tag{FCTP}$$

$$\text{s.t.} \quad -\sum_j x_{ij} \geqslant -s_i, \qquad \forall i \in S \tag{17.91}$$

$$\sum_i x_{ij} \geqslant d_j, \qquad \forall j \in C \tag{17.92}$$

$$-x_{ij} + M_{ij}y_{ij} \geqslant 0 \tag{17.93}$$

$$x_{ij} \geqslant 0, \qquad \forall (i,j) \in A \tag{17.94}$$

$$y_{ij} \in \{0,1\}, \qquad \forall (i,j) \in A \tag{17.95}$$

将主问题变成只有复杂变量 $\boldsymbol{y}$ 的形式，也就是

$$\min_{y} \quad \sum_{(i,j)\in A} f_{ij}y_{ij} + q(\boldsymbol{y}) \tag{FCTP-MP}$$

$$\text{s.t.} \quad y_{ij} \in \{0,1\}, \qquad \forall (i,j) \in A \tag{17.96}$$

Benders 子问题可以写成

$$\min_{x} \quad \sum_{(i,j)\in A} c_{ij}x_{ij} \tag{FCTP-SP}$$

$$\text{s.t.} \quad -\sum_j x_{ij} \geqslant -s_i, \qquad \forall i \in S \tag{17.97}$$

$$\sum_i x_{ij} \geqslant d_j, \qquad \forall j \in C \tag{17.98}$$

$$-x_{ij} + M_{ij}\bar{y}_{ij} \geqslant 0, \qquad \forall (i,j) \in A \tag{17.99}$$

$$x_{ij} \geqslant 0, \qquad \forall (i,j) \in A \tag{17.100}$$

该子问题的对偶为

$$\max_{u,v,w} \quad \sum_i (-s_i)u_i + \sum_j d_j v_j + \sum_{(i,j)\in A} (-M_{ij}\bar{y}_{ij})w_{ij} \tag{FCTP-SP-Dual}$$

$$\text{s.t.} \quad -u_i + v_j - w_{ij} \leqslant c_{ij}, \qquad \forall (i,j) \in A \tag{17.101}$$

$$u_i \geqslant 0, \qquad \forall i \in S \tag{17.102}$$

$$v_j \geqslant 0, \qquad \forall j \in C \tag{17.103}$$

$$w_{ij} \geqslant 0, \qquad \forall (i,j) \in A \tag{17.104}$$

Benders 松弛主问题可以写成

$$\min_{y} \quad \sum_{(i,j)\in A} f_{ij}y_{ij} + z \tag{FCTP-MP-Relaxed}$$

$$z \geqslant \sum_i (-s_i)\,\bar{u}_i^{(k)} + \sum_j d_j \bar{v}_j^{(k)} + \sum_{(i,j)\in A} (-M_{ij}y_{ij})\,\bar{w}_{ij}^{(k)}, \forall k \in K \tag{17.105}$$

$$\sum_i (-s_i)\,\bar{u}_i^{(l)} + \sum_j d_j \bar{v}_j^{(l)} + \sum_{(i,j)\in A} (-M_{ij}y_{ij})\,\bar{w}_{ij}^{(l)} \leqslant 0, \forall l \in L \tag{17.106}$$

$$y_{ij} \in \{0,1\}, z \text{ free} \qquad \forall (i,j) \in A \tag{17.107}$$

其中，(17.105) 就是根据极点添加的 Benders Optimality Cut，而 (17.106) 就是根据极射线添加的 Benders Feasibility Cut。另外，K 和 L 分别为子问题对偶问题（FCTP-MP-Relaxed）的极点和极射线的集合。

上述问题也可以写成下面的等价形式（用 z 替换 $\sum_{(i,j)\in A} f_{ij}y_{ij} + z$）：

$$\min_{y} \quad z \tag{FCTP-MP-Relaxed}$$

$$z \geqslant \sum_{(i,j)\in A} f_{ij}y_{ij} + \sum_i (-s_i)\,\bar{u}_i^{(k)} + \sum_j d_j \bar{v}_j^{(k)} + \sum_{(i,j)\in A} (-M_{ij}y_{ij})\,\bar{w}_{ij}^{(k)}, \forall k \in K \tag{17.108}$$

$$\sum_i (-s_i)\,\bar{u}_i^{(l)} + \sum_j d_j \bar{v}_j^{(l)} + \sum_{(i,j)\in A} (-M_{ij}y_{ij})\,\bar{w}_{ij}^{(l)} \leqslant 0, \qquad \forall l \in L \tag{17.109}$$

$$y_{ij} \in \{0,1\}, z \text{ free} \qquad \forall (i,j) \in A \tag{17.110}$$

本书提供了该案例的完整代码和测试算例，详见附配资源。

17.3.2 设施选址问题

设施选址问题（Facility Location Problem, FLP）可以使用 Benders 分解算法快速求解。首先给出 FLP 的描述（本例中设施为仓库）。

假设可选的仓库点（Candidate Warehouse）的集合为 $W = \{1, 2, \cdots, n\}$，零售店点（retailer）的可选集合为 $R = \{1, 2, \cdots, m\}$。假设开通仓库 i 的固定成本为 f_i，其供货上限为 S_i。如果仓库 i 要为零售店 j 供货，则会产生相应的运输成本，假设单位运输成本为 c_{ij}。我们需要决策在哪些点建设仓库，并且决定仓库与零售店之间的供货关系，使得所有零售店的需求都被满足，同时要最小化总的设施建设成本和服务成本。

基于上述描述，引入下面的决策变量。

- y_i：如果选择在 $i \in W$ 建设仓库，则 $y_i = 1$，否则 $y_i = 0$；
- x_{ij}：仓库 i 向零售店 j 的供货量。

下面给出 FLP 的模型。

$$\min \quad \sum_{i \in W} f_i y_i + \sum_{i \in W} \sum_{j \in R} c_{ij} x_{ij} \tag{17.111}$$

$$\text{s.t.} \quad \sum_{j \in R} x_{ij} \leqslant s_i y_i, \qquad \forall i \in W \tag{17.112}$$

$$\sum_{i \in W} x_{ij} = d_j, \qquad \forall j \in R \tag{17.113}$$

$$y \in \{0,1\}, x_{ij} \geqslant 0, \qquad \forall i \in W, \forall j \in R \tag{17.114}$$

由于仓库选址变量 y_i 为 0-1 变量，在该问题中属于复杂变量，因此可以将其分成主问题和子问题，利用 Benders 分解算法来求解。

主问题中仅有 y_i，具体模型如下。

$$\min_{y} \quad \sum_{i \in W} f_i y_i + q(y) \tag{17.115}$$

$$\text{s.t.} \quad y_i \in \{0,1\}, \qquad \forall i \in W \tag{17.116}$$

求解 (17.115)，得到解 $\bar{y}$（本章代码中，在首次迭代时，将所有 y_i 均初始化为 1，这样方便实现），便可以构建 Benders 子问题，具体模型如下。

$$\min_{x} \quad \sum_{i \in W} \sum_{j \in R} c_{ij} x_{ij} \tag{17.117}$$

$$\text{s.t.} \quad \sum_{j \in R} x_{ij} \leqslant \bar{y}_i s_i, \qquad \forall i \in W \tag{17.118}$$

$$\sum_{i \in W} x_{ij} = d_j, \qquad \forall j \in R \tag{17.119}$$

$$x_{ij} \geqslant 0, \qquad \forall i \in W, \forall j \in R \tag{17.120}$$

模型 (17.117) 的对偶可以写成如下的形式。

$$\max_{\alpha,\beta} \quad \sum_{i \in W} \bar{y}_i s_i \alpha_i + \sum_{j \in R} d_j \beta_j \tag{17.121}$$

$$\text{s.t.} \quad \alpha_i + \beta_j \leqslant c_{ij}, \qquad \forall i \in W, \forall j \in R \tag{17.122}$$

$$\alpha_i \leqslant 0, \qquad \forall i \in W \tag{17.123}$$

$$\beta_j \text{ free}, \qquad \forall j \in R \tag{17.124}$$

因此，Benders 松弛主问题可以写成

$$\min_{y,z} \quad \sum_{i \in W} f_i y_i + z \tag{17.125}$$

$$\text{s.t.} \quad z \geqslant \sum_{i \in W} y_i s_i \bar{\alpha}_i^{(k)} + \sum_{j \in R} d_j \bar{\beta}_j^{(k)}, \qquad \forall k \in K \tag{17.126}$$

$$\sum_{i \in W} y_i s_i \bar{\alpha}_i^{(l)} + \sum_{j \in R} d_j \bar{\beta}_j^{(l)} \leqslant 0, \qquad \forall l \in L \tag{17.127}$$

$$y_i \in \{0,1\}, z \text{ free} \qquad \forall i \in W \tag{17.128}$$

其中，K 和 L 分别为 (17.121) 的极点和极射线的集合。Benders 松弛主问题也可以等价为下面的形式（用 z 替换 $\sum_{(i,j) \in A} f_{ij} y_{ij} + z$）。

$$\min_{y,z} \quad z \tag{17.129}$$

$$\text{s.t.} \quad z \geqslant \sum_{i \in W} f_i y_i + \sum_{i \in W} y_i s_i \bar{\alpha}_i^{(k)} + \sum_{j \in R} d_j \bar{\beta}_j^{(k)}, \qquad \forall k \in K \tag{17.130}$$

$$\sum_{i \in W} y_i s_i \bar{\alpha}_i^{(l)} + \sum_{j \in R} d_j \bar{\beta}_j^{(l)} \leqslant 0, \qquad \forall l \in L \tag{17.131}$$

$$y_i \in \{0,1\}, z \text{ free} \qquad \forall i \in W \tag{17.132}$$

本书提供了本案例的完整代码和测试算例，详见附配资源。

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